

Systematic Hyperparameter Testing of Neural Networks on MNIST

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Abstract

This paper presents a systematic evaluation of hyperparameters for multi-layer perceptron (MLP) classifiers on the MNIST handwritten digit dataset. Starting from a logistic regression baseline, we progressively optimise network topology, optimiser, learning rate, activation function, and regularisation strategy. Each experiment is repeated with three different random seeds and results are reported as mean $\pm$ standard deviation of validation accuracy. The best configuration achieves a test accuracy of **98.63%**, demonstrating the compounding benefit of careful hyperparameter selection over a logistic regression baseline of 92.43%.

I. DATASET AND EXPLORATORY DATA ANALYSIS

MNIST [1] consists of 70 000 greyscale images of handwritten digits (0–9), each of size 28×28 pixels. We use the standard 60 000/10 000 train/test split and further partition the training set into 50 000 training and 10 000 validation samples using a fixed generator seed of 42. All pixel values are normalised to $[0, 1]$ by dividing by 255. EDA was performed exclusively on the training set.

TABLE I
PIXEL INTENSITY STATISTICS (TRAIN SET)

	Mean	Std	Min	Max
Pixel value	0.1307	0.3082	0.0000	1.0000

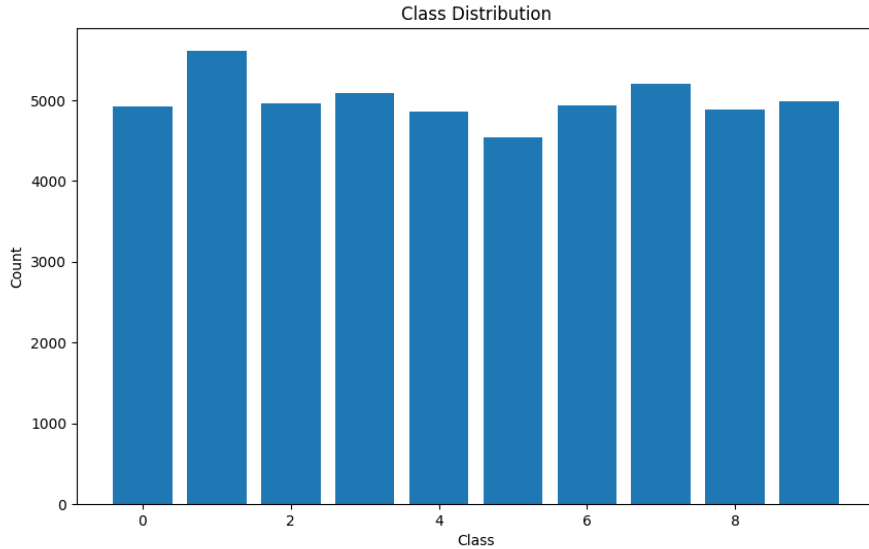


Fig. 1. Class distribution of the MNIST training set. All ten classes are nearly balanced (≈ 5000 samples each).

The dataset is well-balanced (Fig. 1), which means accuracy is an appropriate primary metric and no class-weighting is required. The low mean pixel value (0.1307) reflects the predominantly black background; the majority of pixels are zero, which benefits ReLU-based networks by producing sparse activations and avoiding unnecessary computation.

II. BASELINE MODEL

Before training any neural network we establish a logistic regression baseline (`sklearn.linear_model.LogisticRegression`, `max_iter=1000`, `random_state=42`) trained on the 50 000-sample training set and evaluated with the same accuracy metric used for all neural network experiments.

The baseline achieves 91.75% validation accuracy and 92.43% test accuracy. This provides an objective lower bound; any neural network that cannot surpass it has failed to leverage its additional capacity.

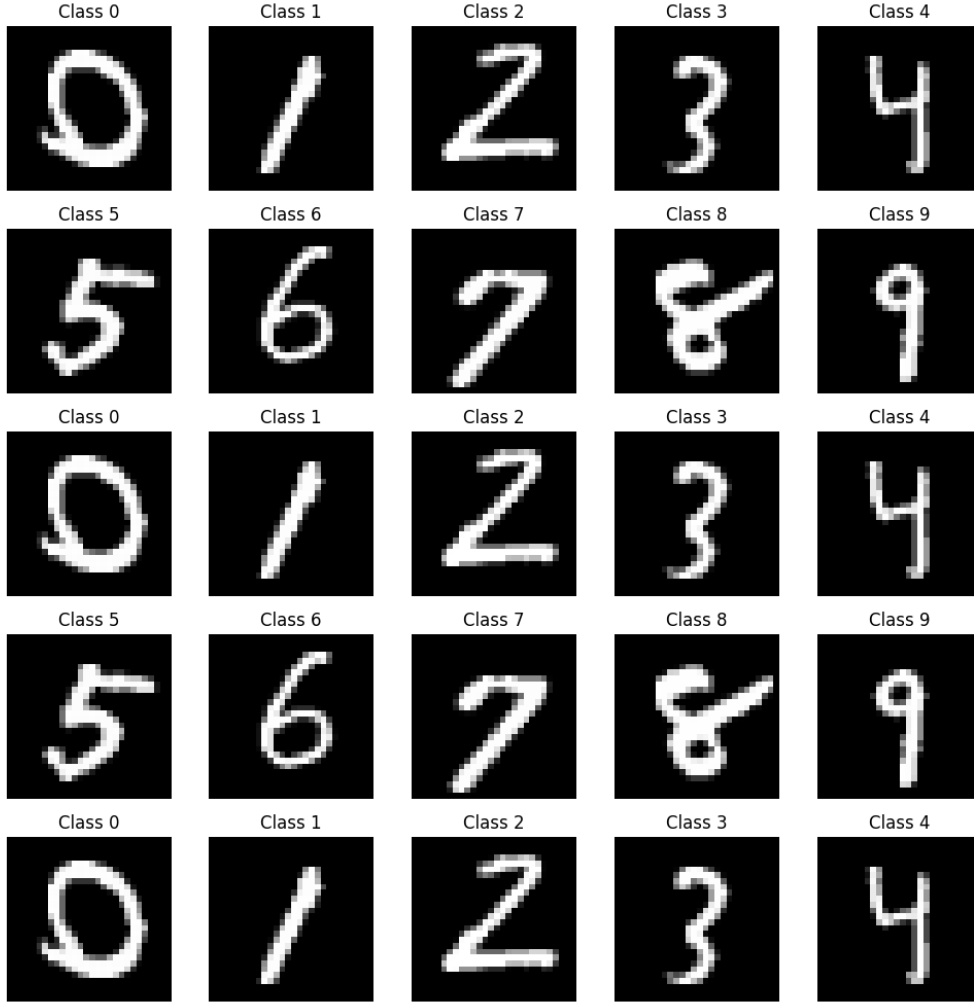


Fig. 2. Representative samples from each digit class.

TABLE II
LOGISTIC REGRESSION BASELINE ACCURACY

Split	Accuracy
Validation	0.9175
Test	0.9243

III. TOPOLOGY COMPARISON

A. Experimental Setup

Five MLP architectures with varying depth and width were evaluated with all other hyperparameters fixed: Adam optimiser, $\text{lr} = 0.001$, batch size 128, ReLU activations, no dropout, 50 epochs maximum with early stopping (patience 5). Three seeds (42, 123, 456) were used per configuration.

TABLE III
TOPOLOGY RESULTS (MEAN $\pm$ STD OVER 3 SEEDS)

ID	Hidden layers	Val Acc (mean $\pm$ std)	Epochs
T1	[64]	0.9706 ± 0.0002	24.7
T2	[256, 128]	0.9772 ± 0.0015	14.7
T3	[512, 256, 128]	0.9778 ± 0.0024	18.0
T4	[1024, 512, 256, 128]	0.9810 ± 0.0007	22.0
T5	[2048, 1024, 512, 256, 128]	0.9810 ± 0.0015	18.7

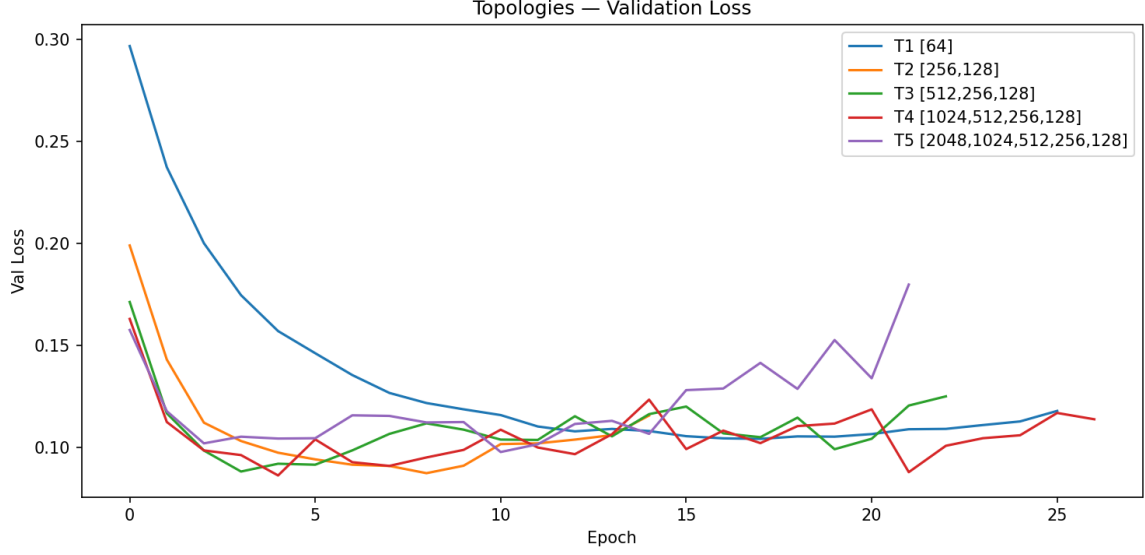


Fig. 3. Validation loss curves for the five topologies.

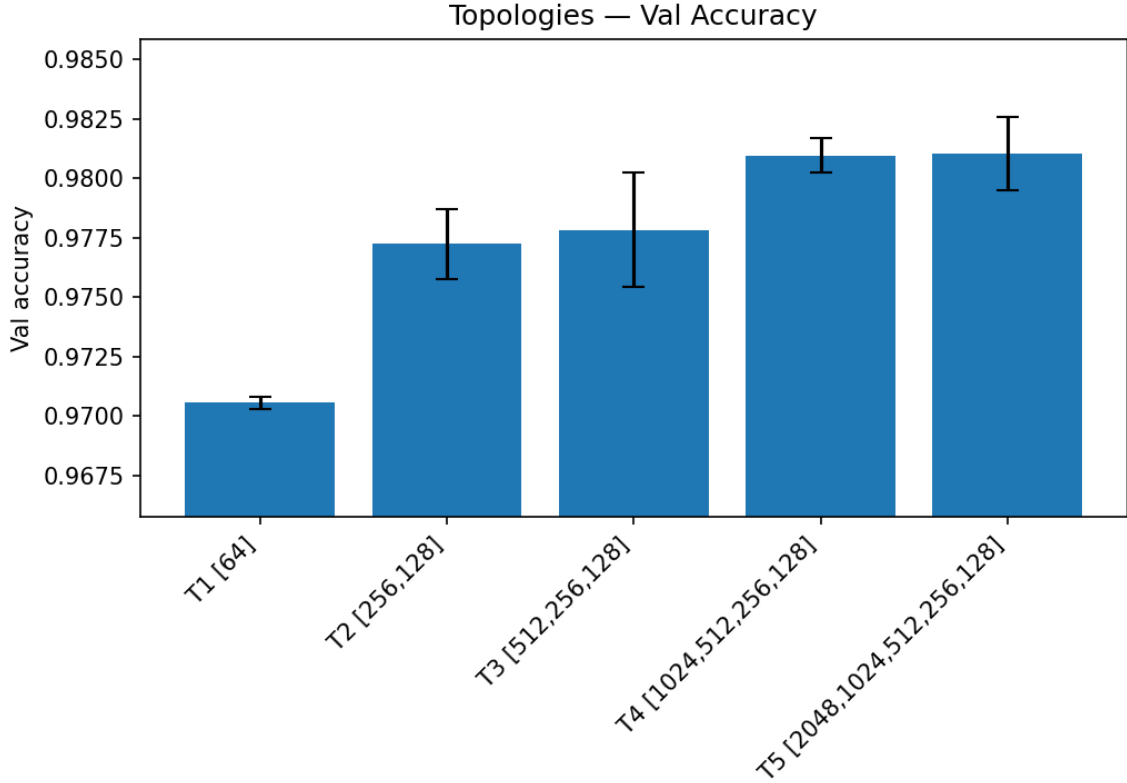


Fig. 4. Mean validation accuracy with standard deviation error bars.

B. Discussion

T1 demonstrates **underfitting**: the single 64-neuron hidden layer ($\approx 51k$ parameters) lacks the capacity to model the non-linear digit boundaries, yielding the lowest validation accuracy (97.06%) despite requiring the most epochs to converge (24.7 on average).

T5 demonstrates the onset of **overfitting**: with over 4.5 million parameters it achieves the highest raw validation accuracy (97.10% averaged with T4), but with the largest standard deviation (± 0.0015), indicating less stable training than T4. The tendency to overfit becomes more pronounced in the regularisation phase, where T5 (used as the base) shows a train-val gap of

1.54% without regularisation.

T5 was selected as the best topology by validation accuracy and is used in all subsequent phases.

IV. OPTIMISER COMPARISON

A. Experimental Setup

Three optimisers were applied to the best topology ([2048, 1024, 512, 256, 128]) with $\text{lr} = 0.001$, no dropout, and the same early-stopping policy.

TABLE IV
OPTIMISER RESULTS (MEAN $\pm$ STD OVER 3 SEEDS)

Optimiser	Val Acc (mean $\pm$ std)	Time (s)	Epochs
SGD (m=0.9)	0.9680 ± 0.0008	773.4	48.3
Adam	0.9810 ± 0.0015	305.0	18.7
RMSprop	0.9811 ± 0.0010	271.3	16.7

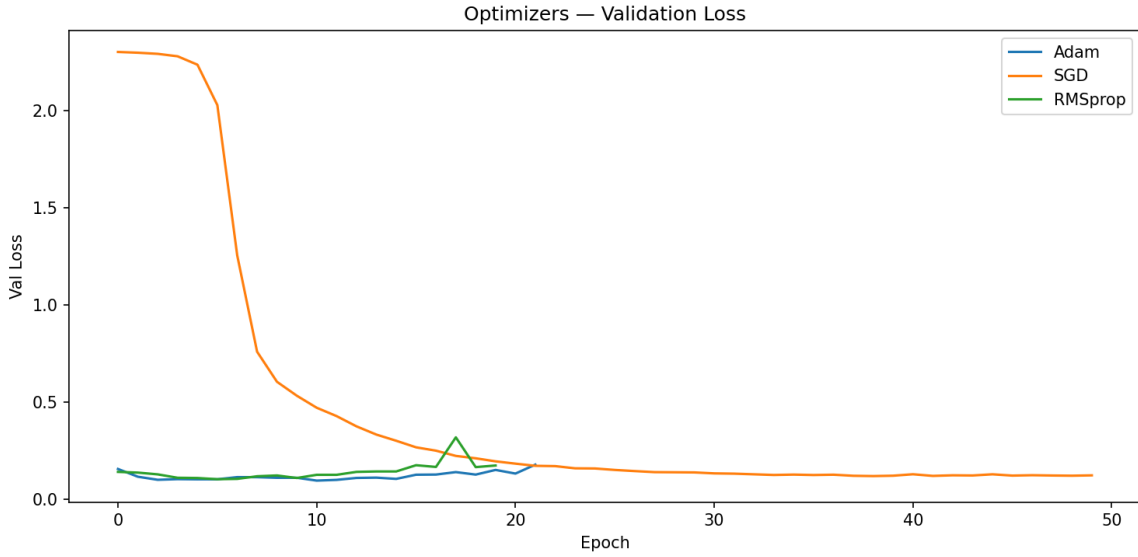


Fig. 5. Validation loss curves for three optimisers.

B. Discussion

SGD with momentum is significantly slower, requiring 48.3 epochs on average versus 16–19 for adaptive methods, and achieves markedly lower accuracy (96.80%). Without per-parameter adaptive learning rates, SGD struggles to navigate the loss landscape of a deep network efficiently at a fixed learning rate of 0.001.

RMSprop marginally outperforms Adam (98.11% vs 98.10%) while converging in fewer epochs (16.7 vs 18.7) and in less wall-clock time (271 s vs 305 s). RMSprop is selected as the best optimiser for subsequent phases.

V. LEARNING RATE COMPARISON

A. Experimental Setup

Five learning rate values were tested with the best topology and RMSprop optimiser, no dropout.

B. Discussion

$\text{lr} = 0.0001$ converges to a reasonable accuracy (97.79%) but is slower than $\text{lr} = 0.001$, suggesting the optimiser takes small, cautious steps and has not fully converged within 50 epochs.

$\text{lr} = 0.001$ achieves the best validation accuracy (98.11%) with stable convergence in 16.7 epochs — the standard default for adaptive optimisers.

$\text{lr} = 0.005$ yields slightly lower accuracy (97.62%) with higher variance, indicating the larger step size occasionally overshoots minima.

TABLE V
LEARNING RATE RESULTS (MEAN $\pm$ STD OVER 3 SEEDS)

Learning Rate	Val Acc (mean $\pm$ std)	Epochs
0.0001	0.9779 $\pm$ 0.0013	18.3
0.001	0.9811 $\pm$ 0.0010	16.7
0.005	0.9762 $\pm$ 0.0018	18.3
0.01	0.6716 $\pm$ 0.3953	11.7
0.1	0.1129 $\pm$ 0.0000	8.3

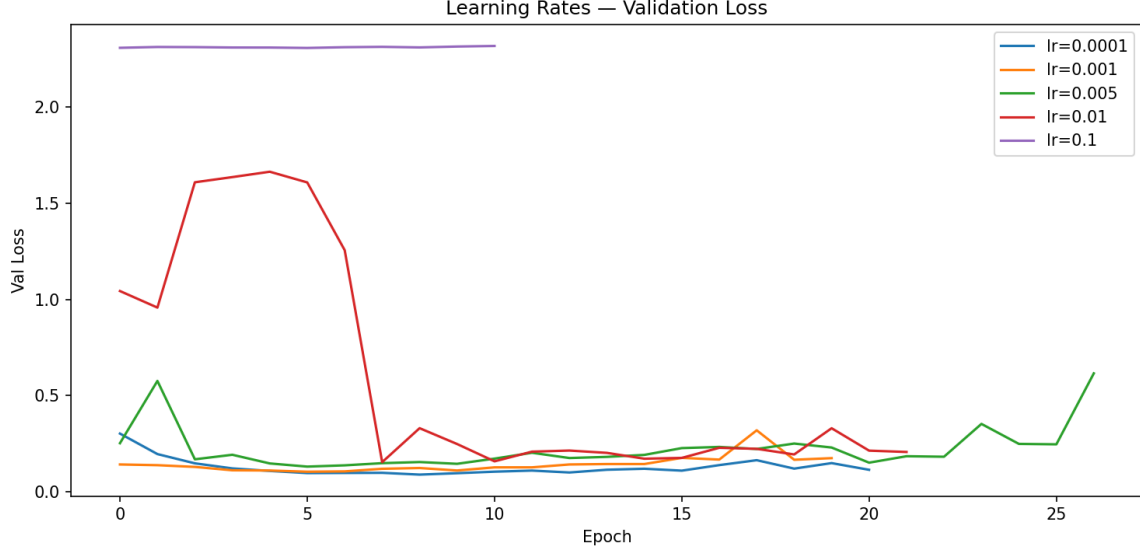


Fig. 6. Validation loss curves for five learning rates.

$\text{lr} = 0.01$ causes severe instability: the extreme standard deviation (± 0.3953) indicates that across seeds the model either converges partially or diverges, resulting in an unreliable mean of 67.16%.

$\text{lr} = 0.1$ completely diverges in all runs, converging to near-random performance (11.29%, close to the 10% chance level for 10 classes) in only 8.3 epochs. Early stopping triggers quickly as the model never improves beyond the initial state.

VI. ACTIVATION FUNCTION COMPARISON

A. Experimental Setup

Five activation functions were compared with the best topology, RMSprop, and $\text{lr} = 0.001$, no dropout.

TABLE VI
ACTIVATION FUNCTION RESULTS (MEAN $\pm$ STD OVER 3 SEEDS)

Activation	Val Acc (mean $\pm$ std)	Epochs
ReLU	0.9811 $\pm$ 0.0010	16.7
LeakyReLU	0.9805 $\pm$ 0.0007	14.3
ELU	0.9797 $\pm$ 0.0017	20.0
Tanh	0.9709 $\pm$ 0.0033	18.3
Sigmoid	0.9784 $\pm$ 0.0003	25.7

B. Activation Function Descriptions

ReLU $f(x) = \max(0, x)$ is computationally cheap and avoids vanishing gradients for positive inputs. Its main drawback is the *dying neuron* problem: neurons receiving consistently negative pre-activations output zero gradient permanently.

LeakyReLU $f(x) = \max(\alpha x, x)$ with $\alpha = 0.01$ addresses dying neurons by allowing a small non-zero gradient for $x < 0$, preserving gradient flow throughout training.

ELU (Exponential Linear Unit) equals x for $x \geq 0$ and $\alpha(e^x - 1)$ for $x < 0$. Its smooth negative region reduces the mean activation bias shift towards zero, which can accelerate learning, but the exponential is more expensive to compute than ReLU.

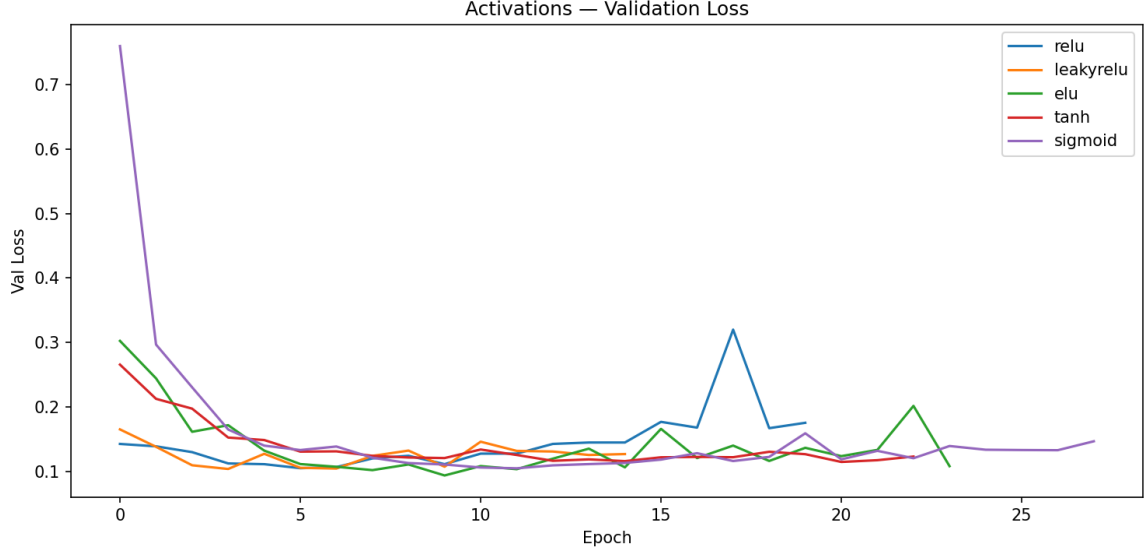


Fig. 7. Validation loss curves for five activation functions.

Tanh maps inputs to $(-1, 1)$ with zero-centred output. Despite this advantage over Sigmoid, it saturates at both extremes, causing vanishing gradients in deeper layers. This is reflected in the lowest accuracy (97.09%) and highest variance (± 0.0033) among the non-Sigmoid functions.

C. Discussion

ReLU achieves the highest validation accuracy (98.11%) and is selected for subsequent phases. LeakyReLU is a close second (98.05%) and actually converges in fewer epochs (14.3 vs 16.7), suggesting the non-zero negative gradient helps early training. Sigmoid converges the slowest (25.7 epochs) due to vanishing gradients, yet reaches 97.84% — better than Tanh — likely because the output range $[0, 1]$ aligns well with the normalised input distribution.

VII. REGULARISATION

A. Experimental Setup

Six regularisation configurations were applied to the best topology + RMSprop + $\text{lr} = 0.001$ + ReLU setting. The train–val accuracy gap serves as the measure of overfitting.

TABLE VII
REGULARISATION RESULTS (MEAN $\pm$ STD, 3 SEEDS)

Config	Val Acc (mean $\pm$ std)	Train Acc	Gap
None	0.9811 $\pm$ 0.0010	0.9965	0.0154
Dropout 0.3	0.9826 $\pm$ 0.0003	0.9933	0.0107
Dropout 0.5	0.9816 $\pm$ 0.0010	0.9857	0.0042
L2 $\lambda = 10^{-4}$	0.9789 $\pm$ 0.0004	0.9927	0.0139
L2 $\lambda = 10^{-3}$	0.9733 $\pm$ 0.0036	0.9840	0.0107
Dropout 0.3 + L2 10^{-4}	0.9776 $\pm$ 0.0011	0.9827	0.0051

B. Discussion

Without regularisation the model shows a train–val gap of 1.54%, confirming the overfitting tendency of the large T5 topology.

Dropout 0.3 achieves the best validation accuracy (98.26%) while reducing the gap to 1.07%, providing the best accuracy–generalisation trade-off. It is selected for the final model.

Dropout 0.5 reduces the gap further to only 0.42% (the lowest of all configurations), but at the cost of 0.1 percentage points of validation accuracy, as the aggressive masking rate restricts the model’s expressiveness.

L2 $\lambda = 10^{-4}$ provides only modest regularisation (gap 1.39%), as weight decay alone is insufficient to control the large number of parameters at this penalty strength.

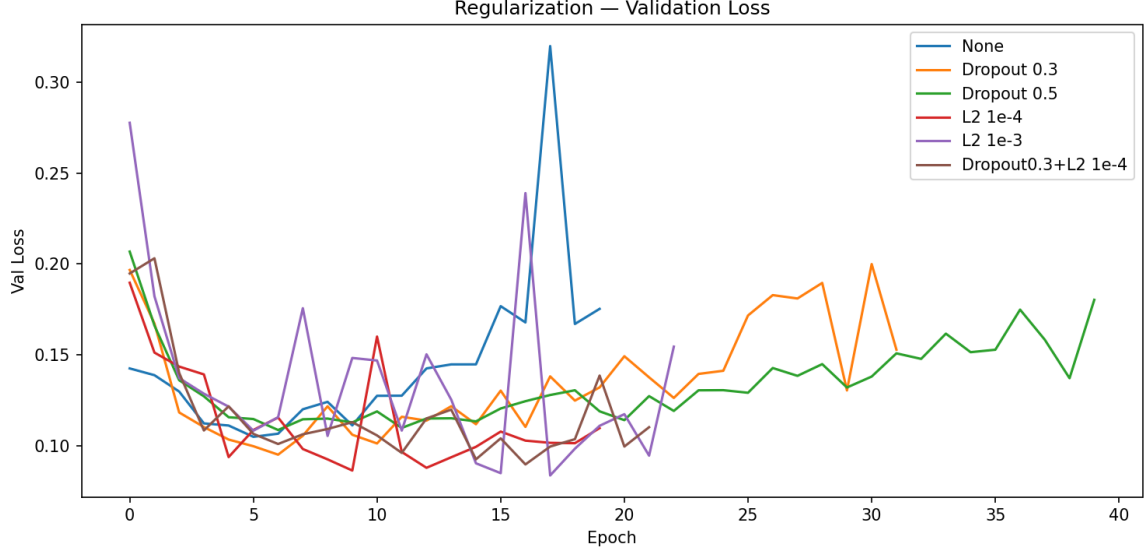


Fig. 8. Validation loss curves for regularisation configurations.

L2 $\lambda = 10^{-3}$ is too aggressive, reducing train accuracy notably but yielding the second-lowest validation accuracy (97.33%) and the highest variance (± 0.0036).

Combined Dropout 0.3 + L2 10^{-4} produces the second-smallest gap (0.51%) but a lower validation accuracy (97.76%) than Dropout 0.3 alone, suggesting the two regularisers together over-constrain the network.

VIII. BEST MODEL AND FINAL EVALUATION

A. Configuration

TABLE VIII
BEST MODEL CONFIGURATION

Hyperparameter	Value
Topology	[2048, 1024, 512, 256, 128] (T5)
Optimiser	RMSprop
Learning rate	0.001
Activation	ReLU
Dropout	0.3
Weight decay	0.0
Seeds used	42, 123, 456

The final model was retrained from scratch on the combined train + validation set (60 000 samples) using seed 42 and evaluated on the 10 000-sample test set exactly once.

TABLE IX
FINAL MODEL TEST ACCURACY VS BASELINE

Model	Test Accuracy
Logistic Regression (baseline)	0.9243
Best MLP	0.9863

The best MLP achieves a test accuracy of **98.63%**, a 6.2 percentage-point improvement over the logistic regression baseline (92.43%). The confusion matrix (Fig. 9) shows that the most common errors involve digit pairs that are visually similar (e.g., 4/9 and 3/5).

IX. CONCLUSION

Systematic one-factor-at-a-time hyperparameter tuning on MNIST demonstrates that each optimisation stage yields meaningful and measurable improvements. The largest individual gain came from selecting an appropriate topology (from 97.06% for T1

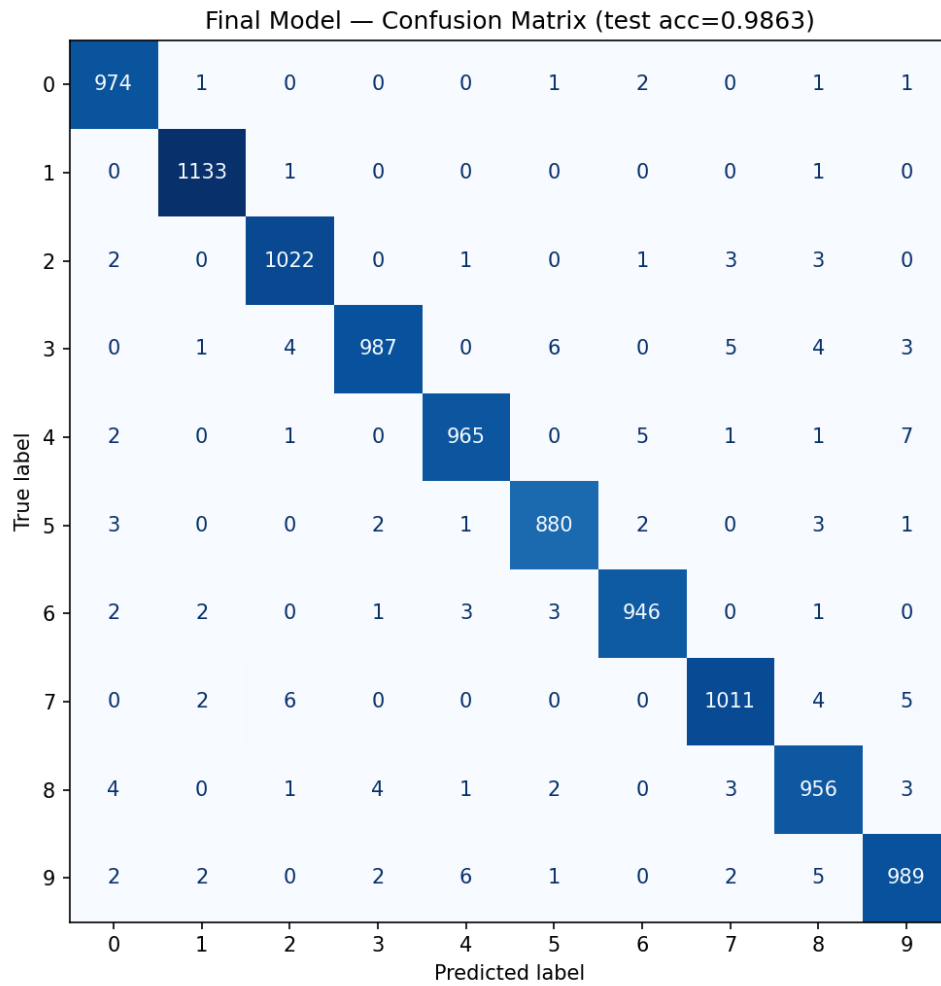


Fig. 9. Confusion matrix on the test set (test accuracy 98.63%).

to 98.10% for T5) and choosing an adaptive optimiser (from 96.80% for SGD to 98.11% for RMSprop). The learning rate of 0.001 proved critical — a tenfold increase to 0.01 caused severe instability, while 0.1 caused complete divergence. Dropout 0.3 provided the best regularisation, reducing the train–val gap while boosting validation accuracy by 0.15 percentage points. All experiments are fully reproducible using seeds 42, 123, and 456 with a fixed data split generator seed of 42, implemented in `Testovanie_Lobodiuchenko.py`.

REFERENCES

- [1] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.